

One
Di-
men-
sion
Two
Di-
men-
sions
 x_2
Three,
Four
po-
lar
co-
or-
di-
nate.

P
 (r, θ)
 r
signed
di-
rected
 P
 θ
 P

Positive
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mea-
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mea-
sured
coun-
ter-
clock-
wise
from
the
ori-
gin.

2π
 (r, θ)
 $(-r, \theta +$
 $\pi)$
 $(-r, \theta -$
 $\pi)$
 $P(r, \theta), r >$

0
 θ
 $(r, \theta +$
 $2\pi)$
 (r, θ)
 $P(r, \theta)$
 $P'(-r, \theta)$

$Q(1, \frac{5\pi}{3})$
 $R(-2, \frac{3\pi}{2})$
 $S(3, -\frac{9\pi}{4})$
 $S'(3, \frac{1\pi}{4})$
 $S''(-3, -\frac{5\pi}{4})$

$\pi/2/above, 4/\pi/left, 6/3\pi/2/below(0, 0) - (*180/4 : 4.1); [fill = white]at(*180/4 : 4.2)[];$

$Q(1, \frac{5\pi}{3})$
 $R(-2, \frac{3\pi}{2})$
 $S(3, -\frac{9\pi}{4})$

θ
 $P(x, y) =$
 $P(r, \theta)$
 (r, θ)
 (x, y)

$x = r \cos \theta$ and $y = r \sin \theta$

(1)

(x, y)
 (r, θ)

$r^2 = x^2 + y^2$ and $\tan \theta = \frac{y}{x}$

(2)

BE
SURE
TO
GRAPH
POINTS
IN
CARTE-
SIAN